Differentiation

Questions 1–112

1.30 — Question 1. For $f(x) = 2x^2 + 1$, find

a) $f(1)$
b) $f(-1)$
c) $f(3)$
d) $f(-4)$
e) $f\left(\frac{1}{2}\right)$

1.30 — Question 2. For each of the following curves, the points A and B lie on the curve when $x = 0$ and $x = 1$ respectively. Find the gradient between these two points on:

a) $y = 3x^2 + x + 1$
b) $y = 2^x + 1$
c) $y = \frac{1}{x - 2}$
d) $y = \frac{x}{1 + x}$

1.30 — Question 3. The table below shows some calculations made with the curve $f(x) = 7x^2$.

x	$f(x)$	h	$x+h$	$f(x+h)$
2	28	1	3	63
2	28	0.1	2.1	30.87
2	28	0.01	2.01	28.2807
2	28	0.001	2.001	
2	28	0.0001		

Complete the table and hence state the value of $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$.

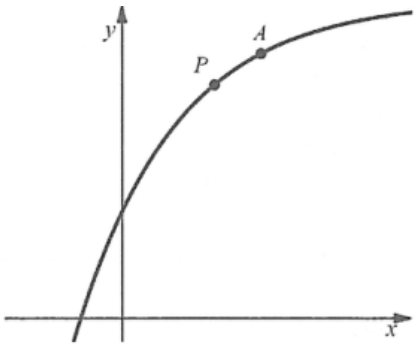
1.30 — Question 4. The table below shows some calculations made with the curve $y = f(x)$.

x	$f(x)$	h	$x+h$	$f(x+h)$
1	4	1	2	6
1	4	0.1	1.1	4.143546925
1	4	0.01	1.01	4.0139111
1	4	0.001	1.001	4.001386775
1	4	0.0001	1.0001	4.000138634

a) Calculate $\dfrac{f(x+h)-f(x)}{h}$ for each line of the table, giving your answers to 5 decimal places.

b) State the value of $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$ to 2 decimal places.

1.30 — Question 5. The diagram below shows $y = -5 \times (0.4)^x + 10$.



The point P lies on the curve when $x = 2$ and the point A lies on the curve when $x = 3$.

a) Find the gradient of the chord connecting the two points A and P.

A third point B lies on the curve when $x = 2.1$.

b) Find the gradient of the chord connecting the two points B and P giving your answer to 3 significant figures.

c) Explain how you could improve on the approximation to the gradient of the tangent at the point P.

1.30 — Question 6. A curve has equation $y = \dfrac{1}{x-3}$. The point $P(2, -1)$ lies on the curve. Find the gradient of the chord connecting P with each of the following values of x and where necessary explain why this does not approximate the gradient at $x = 2$:

a) $x = 2.5$

b) $x = 3$

c) $x = 3.5$

1.30 — Question 7. Expand the following:

a) $(1+h)^2$
b) $5(1+h)^2$
c) $(1+h)^3$
d) $-3(1+h)^3$

1.30 — Question 8. Expand the following:

a) $(2+h)^2$
b) $7(3+h)^2$
c) $(-2+h)^3$
d) $-3(10+h)^3$

1.30 — Question 9. Expand and simplify the following:

a) $(3+h)^2 - 9$
b) $6(1+h)^2 - 6$
c) $(-1+h)^3 + 1$
d) $-2(4+h)^3 + 128$

1.30 — Question 10. Find $f(2)$ and $f(2+h)$ for each of the following:

a) $f(x) = x^2$
b) $f(x) = x^2 - x$
c) $f(x) = x^2 - 2x + 5$
d) $f(x) = x - 3x^2$
e) $f(x) = 10 - x - 2x^2$

1.30 — Question 11. Find the value of each of the following:

a) $\lim_{h\to 0}(2+h)$

b) $\lim_{h\to 0}(h-5)$

c) $\lim_{h\to 0}(2h+3)$

d) $\lim_{h\to 0}\left(\frac{5h^2+h}{h}\right)$

e) $\lim_{h\to 0}\left(\frac{6h^2-4h}{h}\right)$

1.30 — Question 12. Explain why $\lim_{h\to 0}\left(\frac{h^2-2}{h}\right)$ cannot be found.

1.30 — Question 13.

a) Expand $(1+h)^3$.

b) Simplify $\dfrac{8(1+h)^3-8}{h}$.

c) Hence find the gradient of the curve $y=8x^3$ when $x=1$.

1.30 — Question 14. Using differentiation from first principles, find the gradient of the curve of $f(x)$ at the point given.

a) $f(x)=x^2$ at $x=2$

b) $f(x)=3x^2$ at $x=1$

c) $f(x)=x^2-5x$ at $x=-1$

1.30 — Question 15. A curve is given as $f(x) = x^2 + 5$.

a) Calculate the gradient of the chord joining the two points on the curve when $x = 4$ and $x = 4.1$.

b) Find, in its simplest form, $\dfrac{f(4+h) - f(4)}{h}$.

c) Hence, state the gradient of $f(x)$ when $x = 4$.

1.30 — Question 16. The point $(3, 45)$ lies on the curve $y = 5x^2$ and at this point has the gradient m.

a) Show that $m = \lim_{h \to 0} (30 + 5h)$.

b) Deduce the value of m.

1.30 — Question 17. Using differentiation from first principles, show that the derivative of $5x^2$ is $10x$.

1.30 — Question 18. Differentiate from first principles:

a) $y = 2x^2 + 1$
b) $y = 5x^2 + x$
c) $y = x - 4x^2$
d) $y = x^2 - 2x + 1$
e) $y = x^3$

1.30 — Question 19. Use differentiation from first principles to find the exact value of $f'(2)$ for each of the following:

a) $f(x) = 3x^2$

b) $f(x) = 1 - 2x^2$

1.30 — Question 20. Show from first principles that the gradient function of kx^3 is $3kx^2$ where k is a constant.

1.30 — Question 21. A curve has equation $y = 12x - x^3$. It passes through the point $P(2, 16)$.

a) Find the gradient between point P and point Q which has x-ordinate $2 + h$.

b) Using your answer to part a), find the gradient at the point P.

1.30 — Question 22. James is attempting to use differentiation from first principles to prove that the gradient of $f(x) = 2x^3 - 9$ at $x = 3$ is 54. Find the two mistakes in his workings below and correct his proof:

$$f(3) = 2 \times 3^3 - 9 = 45$$

$$f(3 + h) = 2(3 + h)^3 - 9 = 2h^3 + 9h^2 + 54h + 45$$

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{2h^3 + 9h^2 + 54h + 45 - 45}{h}$$

$$= 2h^2 + 9h + 54 = 54$$

1.30 — Question 23. Using differentiation from first principles find the derivative of $y = ax + b$, where a and b are constants.

1.30 — Question 24. Using differentiation from first principles find the derivative of $y = ax^2 + bx + c$, where a, b and c are constants.

1.30 — Question 25. When $x = a$, where $a > 3$, the curve $y = f(x)$ is 8 units higher than when $x = 3$. Given that $f(x) = x^2 + 2x - 7$, find the exact value of a.

1.30 — Question 26. It is known that two functions $f(x) = px^3$ and $g(x) = qx^2 - 5px + 1$ are related by $f'(2) = g'(2) = 24$. By using differentiation from first principles, find the values of p and q.

1.30 — Question 27.

a) Using differentiation from first principles, find the slope of the tangent to the curve $f(x) = 4x^2 - 7$ at the point $P(2, 9)$.

b) Calculate the positive angle which the tangent makes with the positive x-axis at P giving your answer in degrees to 1 decimal place.

1.30 — Question 28. Using differentiation from first principles, for the curve $y = 4x^2 - x + 1$ find:

a) The gradient of the curve when $x = 1$.

b) The point on the curve where the gradient is 15.

1.30 — Question 29. Use differentiation from first principles to find $\frac{dy}{dx}$ for each of the following:

a) $y = \frac{1}{x}$

b) $y = \frac{1}{1-x}$

c) $y = \sqrt{x+1}$

1.30 — Question 30. By differentiating $f(x) = \frac{1}{7x+2}$, $x \neq -\frac{2}{7}$ from first principles show that $f'(x) = -\frac{7}{(7x+2)^2}$.

1.30 — Question 31. By differentiating $f(x) = \dfrac{2}{3x-1}$, $x \neq \dfrac{1}{3}$ from first principles show that $f'(x) = -\dfrac{6}{(3x-1)^2}$.

1.30 — Question 32. Prerequisite knowledge for Q32: Stationary points Two points A and B lie on curve with equation $y = x^3 - 12x$. Point A is when $x = 2$ and point B is when $x = 2 + h$.

a) Show that the gradient of the chord AB is $h^2 + 6h$.

b) How does your answer to part a) demonstrate point A is a stationary point?

1.30 — Question 33. Prerequisite knowledge for Q33–Q34: Binomial Expansion Differentiate from first principles:

a) $y = x^5$

b) $y = 2x^4$

1.30 — Question 34. Use differentiation from first principles to show that

a) $\frac{d}{dx}[(1-x)^3] = -3(1-x)^2$

b) $\frac{d}{dx}[(2-x)^3] = -3(2-x)^2$

c) $\frac{d}{dx}[(1-2x)^3] = -6(1-2x)^2$

1.31 — Question 1. Find $\frac{dy}{dx}$ for each of these:

a) $y = 4x^3 + 5x^2$
b) $y = 2x - 9x^2$
c) $y = x^6 - 18x^2 + 3$
d) $y = x^{-1} + x^4$
e) $y = x^3 - x^{-2}$
f) $y = x^2 + x^{-2} + x + 1$
g) $y = 3x^5 - 2x^{-2} - x + 6$
h) $y = 2x^a - x^b$

1.31 — Question 2. Find $f'(t)$ for each of these:

a) $f(t) = \frac{1}{2}t^2 + \frac{3}{5}t^{-5}$
b) $f(t) = \frac{1}{4}t - \frac{3}{4}t^4$
c) $f(t) = \frac{1}{12}t^6 + 10t^2 + 7$
d) $f(t) = \frac{1}{3}t^{-1} + \frac{2}{3}t^6$
e) $f(t) = \frac{7}{8}t^2 - \frac{1}{4}t^{-3} - 4$
f) $f(t) = \frac{2}{7}t^8 + \frac{1}{7}t^{-3} + \frac{1}{5}t + \frac{4}{7}$
g) $f(t) = -\frac{1}{4}t^6 - \frac{1}{2}t^{-2} + 2t + \frac{5}{9}$
h) $f(t) = -\frac{1}{2}t^{a/b}$, $b \neq 0$

1.31 — Question 3. Find $\frac{dy}{dt}$ for each of these:

a) $y = 8t^{1/2} + 6t^{1/3}$
b) $y = 5t - 12t^{1/4}$
c) $y = t^{2/3} + 7t^{3/5}$
d) $y = t^{-1/2} + t^{1/4} + \frac{1}{4}$
e) $y = -t^{5/8} - t^{-3/8} - \frac{1}{8}t + \frac{1}{8}$
f) $y = t^{1/3} + t^{-1/3} + 3t + 3$
g) $y = -t^{3/4} + t^{-8/9} - \frac{1}{12}t + 1$

1.31 — Question 4. Find $f'(x)$ for each of these:

a) $f(x) = x(x+3)$
b) $f(x) = (x+1)(x+5)$
c) $f(x) = (x-2)(2x+5)$
d) $f(x) = (x+4)^2$
e) $f(x) = x^2(x-1)$
f) $f(x) = (x+2)^2(5x-1)$
g) $f(x) = x(7x - x^3)$
h) $f(x) = x^2(9x - x^{-1})$
i) $f(x) = x(7x^{-1} - x^{-2} + 8)$
j) $f(x) = \left(x + \frac{1}{x}\right)^3$
k) $f(x) = (x+p)(x+q)$
l) $f(x) = (x-a)\left(x + \frac{1}{a}\right)$, $a \neq 0$

1.31 — Question 5. Write each of the following in the form x^k:

a) $\sqrt{x}$
b) $\sqrt{x^5}$
c) $\dfrac{1}{x^4}$
d) $(x\sqrt{x})^3$
e) $\sqrt[3]{x}$
f) $\dfrac{1}{\sqrt{x}}$
g) $\dfrac{x}{\sqrt{x}}$
h) $\dfrac{1}{x\sqrt{x}}$
i) $\dfrac{1}{x \times \sqrt[a]{x}}$ where $a > 0$

1.31 — Question 6. Write each of the following in the form ax^b:

a) $5\sqrt[3]{x}$
b) $\sqrt{4x^3}$
c) $\dfrac{3}{x^2}$
d) $(2x\sqrt{x})^5$
e) $\dfrac{\sqrt[8]{x}}{3}$
f) $\dfrac{4}{\sqrt{x}}$
g) $\dfrac{3x}{\sqrt{9x^4}}$
h) $\dfrac{5}{x\sqrt{16x}}$
i) $\dfrac{2}{\sqrt{25ax}}$ where $a > 0$

1.31 — Question 7. Find $\frac{dy}{dx}$ for each of these:

a) $y = \sqrt[4]{x}$
b) $y = \sqrt{9x^5}$
c) $y = \frac{5}{x^3}$
d) $y = (x\sqrt{x})^3$
e) $y = \frac{\sqrt[3]{x}}{5}$
f) $y = \frac{3}{\sqrt{x}}$
g) $y = \frac{x}{\sqrt{x^4}}$
h) $y = \frac{2x}{\sqrt{16x^4}}$

1.31 — Question 8. Find $f'(x)$ for each of these:

a) $f(x) = \sqrt{x}(x+2)$
b) $f(x) = \sqrt{x}(x^2-1)$
c) $f(x) = \frac{x^3+4x}{x}$
d) $f(x) = \frac{2x^3+4x^{-2}}{x}$
e) $f(x) = \frac{9x-1}{x^2}$
f) $f(x) = \frac{x^2+2x-1}{\sqrt{x}}$
g) $f(x) = \frac{x^2-5x^{-3/2}+1}{\sqrt{x}}$

1.31 — Question 9. For each of the following, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$:

a) $y = x^3 + x^{-3} + 2x^2 - 5x + 8$

b) $y = \frac{1}{12}x^4 + \frac{1}{8}x^2 + 7x$

c) $y = -x^3 - 2x^{-1/2} + 15x$

d) $y = -x^{1/7} - x^{-1/7} - \frac{1}{9}x + \frac{1}{9}$

e) $y = x^3(2x + x^{-2})$

f) $y = \sqrt{16x^3}$

g) $y = \frac{2}{\sqrt{x}}$

h) $y = \frac{x^5 + 4x^{-2}}{x}$

1.31 — Question 10. Find the value of the gradient of the curve at the given points:

a) $y = x^2 + 5x - 1$ at $x = 1$

b) $y = x^3 - 2x + 4$ at $x = -2$

c) $y = \frac{1}{\sqrt{x}} - 2x^2$ at $x = 4$

d) $y = \sqrt{x} - x$ at $(4, -2)$

e) $y = \frac{x^4 - 3x^3}{x}$ at $(2, -4)$

1.31 — Question 11. Find $f''(1)$ for each of these:

a) $f(x) = x^3 - 5x^2 + 2$

b) $f(x) = \dfrac{1}{x^2} - x^2$

c) $f(x) = \sqrt{x} + 2x^2$

d) $f(x) = x\sqrt[3]{x} - 4x$

e) $f(x) = p\sqrt{x} - \dfrac{4}{qx}$ where $q \neq 0$

1.31 — Question 12. Prerequisite knowledge for Q12–Q17: Differentiation of the form ae^{kx}

Find $f'(x)$ for each of these:

a) $f(x) = e^{2x}$

b) $f(x) = e^{13x}$

c) $f(x) = e^{-8x}$

d) $f(x) = e^{x/5}$

e) $f(x) = e^{-4x/7}$

f) $f(x) = 2e^{3x}$

g) $f(x) = 5e^{-x}$

h) $f(x) = \dfrac{1}{4}e^{-2x}$

i) $f(x) = \dfrac{1}{e^{5x}}$

j) $f(x) = \sqrt{e^{3x}}$

k) $f(x) = \dfrac{e^x}{\sqrt{e^x}}$

l) $f(x) = \dfrac{e^x}{\sqrt{e^{6x}}}$

m) $f(x) = \sqrt{e^{ax}}$

n) $f(x) = \dfrac{e^x}{\sqrt{e^{ax}}}$

1.31 — Question 13. Find $\frac{dy}{dx}$ for each of these:

a) $y = 2e^{2x} - 5e^{3x}$
b) $y = 2x - e^{3x}$
c) $y = e^{-x} + x^5$
d) $y = x^5 - \frac{1}{4}x^4 + 3e^{-x/5}$
e) $y = \sqrt{x} + \frac{3}{e^{2x}}$
f) $y = \sqrt{ax} + \frac{5}{e^{bx}}$

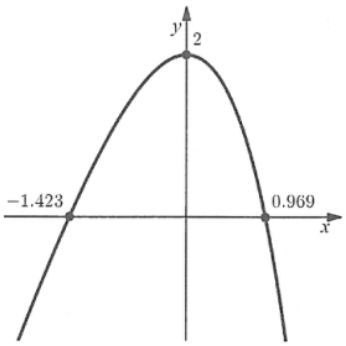
1.31 — Question 14. Find $f'(x)$ and $f''(x)$ for each of these:

a) $f(x) = e^{3x} + 8e^{-x}$
b) $f(x) = 5x^2 - e^{5x}$
c) $f(x) = 2e^{-x} + x^2$
d) $f(x) = x^4 - 8e^{-x/2} - \frac{1}{2}x^{-3}$

1.31 — Question 15. Find the value of the gradient of the curve at the given points, giving your answers in exact form:

a) $y = e^x$ at $x = 2$
b) $y = 5e^x$ at $x = -1$
c) $y = x^2 - e^x$ at $(0, -1)$
d) $y = x - e^x$ at $(1, 1 - e)$
e) $y = 2(x + e^{2x})$ at $\left(\frac{1}{2}, 1 + 2e\right)$
f) $y = -3(x + e^{ax})$ at $(1, -3 - 3e^a)$

1.31 — Question 16. The graph below shows the curve $y = 3(x - e^x) + 5$. Find the value of the gradient at each of the points where the curve meets the axes giving your answers to 2 decimal places.



1.31 — Question 17. Find the values of k and p such that the second derivative of $y = \dfrac{k}{e^{px}}$ is always positive.

1.32 — Question 1. Evaluate $f'(2)$ for each of the following, leaving your answers in exact form:

a) $f(x) = x^3 - 2x^{-1}$

b) $f(x) = \sqrt{x} + 2x^4$

c) $f(x) = \dfrac{x^5 - 1}{x^2}$

d) $f(x) = \sqrt{x}(1 - x)$

1.32 — Question 2. For each of the following curves, find the equation of the tangent line which passes through the given point. Give your answers in the form $ax + by + c = 0$ where a, b, c are integers and $a > 0$:

a) $y = x^2$ at $(2, 4)$

b) $y = x^3 - x + 5$ at $(1, 5)$

c) $y = \dfrac{1}{x}$ at $\left(-2, -\dfrac{1}{2}\right)$

d) $y = 2\sqrt{x}$ at $x = 1$

1.32 — Question 3.

a) Show that the tangent to the curve $y = 3 - x^2$ when $x = 0$ may be written in the form $y - k = 0$ where k is a value to be found.

b) Explain the significance of the gradient of this tangent.

1.32 — Question 4. For each of the following curves, find the equation of the normal line which passes through the given point. Give your answers in the form $ax + by + c = 0$ where a, b, c are integers and $a > 0$:

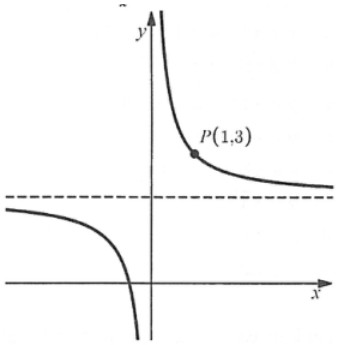
a) $y = 3x^2$ at $(1, 3)$
b) $y = 5x^4 - x^2$ at $(-1, 4)$
c) $y = \dfrac{1}{x^2}$ at $\left(2, \dfrac{1}{4}\right)$
d) $y = \sqrt{x}$ at $(1, 1)$
e) $y = x - x^4$ when $x = -1$

1.32 — Question 5. For the equation $y = 5x^2 + 1$, find the value of x such that $\dfrac{dy}{dx} = 30$.

1.32 — Question 6. For the equation $y = x^3 - 2x$, find the values of x such that $\dfrac{dy}{dx} = 10$.

1.32 — Question 7. For the equation $y = x^{1/2}$, find the value of x such that $\dfrac{dy}{dx} = \dfrac{1}{4\sqrt{2}}$.

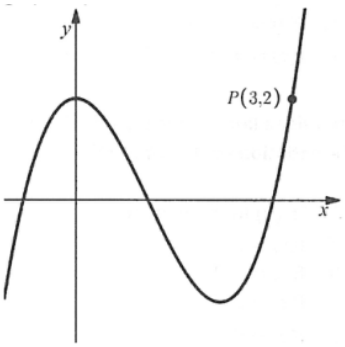
1.32 — Question 8. The graph of $y = \dfrac{1}{x} + 2$ is shown below.



The point P has coordinates $(1, 3)$.

a) Find the equation of the tangent line at P in the form $y = mx + c$ where m and c are integers to be found.
b) Find the equation of the normal line at P in the form $y = mx + c$ where m and c are integers to be found.
c) Find the coordinates where the tangent and normal line at P meets the coordinate axes.

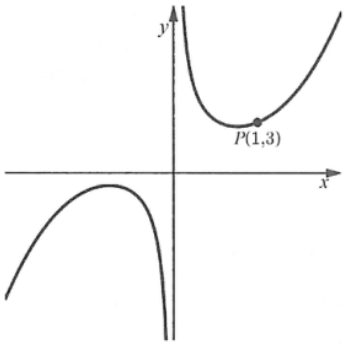
1.32 — Question 9. The graph of $y = x^3 - 3x^2 + 2$ is shown below.



The point P has coordinates $(3, 2)$.

a) Find the equation of the normal line at P in the form $ax + by = c$ where a, b and c are integers to be found and $a > 0$.
b) Find the coordinates where the normal line crosses the x-axis and y-axis.
c) Hence calculate the area of the triangle bounded by the normal line at P and the coordinate axes.

1.32 — Question 10. The graph of $y = \dfrac{1}{x} + x^3 + 1$ is shown below.



The point P has coordinates $(1, 3)$.

a) Find the equation of the tangent line at P in the form $y = mx + c$ where m and c are integers to be found.
b) Find the coordinates where the tangent line crosses the x-axis and y-axis.
c) Find the coordinates where the tangent meets the curve again.

1.32 — Question 11. The point A lies on the curve

$$y = 5x^2 - \frac{1}{x}.$$

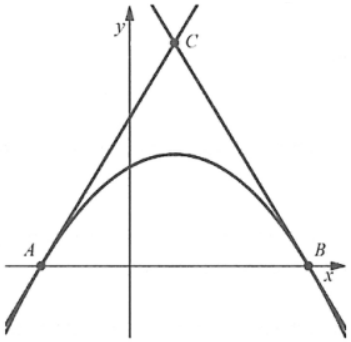
The tangent line to the curve at point A which lies on the curve is parallel to the line $y + 9x = 5$.

a) Find the coordinates of the point A.
b) Find the equation of the tangent line to the curve at the point A in the form $y = mx + c$ where m and c are integers to be found.
c) Hence find the other point where the tangent intersects the curve.

1.32 — Question 12. A curve has equation $y = \frac{k}{x}$ where k is a positive constant. Find $\frac{dy}{dx}$ at $\left(\frac{1}{2}, 2k\right)$ giving your answer in terms of k.

1.32 — Question 13. A curve has equation $y = x^3 + px^2 + qx + 5$. It passes through the point $P(2, 1)$. The gradient of the curve at P is 10. Find the value of p and q.

1.32 — Question 14. The graph below shows the curve $f(x) = -x^2 + 2x + 8$ which meets the x-axis at points A and B. The tangent lines to the curve at A and B meet at point C. Find the area of the triangle ABC.



1.32 — Question 15. A curve has equation $y = x^3 + (1 + p)x^2 + qx + 5$. It passes through the point $P(-1, 6)$. The gradient of the normal to the curve at P is $\frac{1}{2}$. Find the value of p and q.

1.32 — Question 16. A curve has equation $y = 3x^3 - 5x^5 + 1$. At $x = -1$ the tangent line has the same gradient as another tangent line to the curve. Find the equation of this other tangent line.

1.32 — Question 17. Show that the equation of the tangent to the curve $y = \dfrac{1}{x^2}$ at the point $x = a$ is given by $a^3y + 2x = 3a$.

1.32 — Question 18.

a) Show that the equation of the tangent to the curve $y = x(x^2+2)$ at the point $x = a$ is given by $y = (3a^2+2)x-2a^3$.

b) Hence explain why the gradient of the curve $y = x(x^2 + 2)$ is always positive.

1.32 — Question 19. Prerequisite knowledge for Q19: Second derivatives A curve has equation $y = x^2(1 - kx)$ where k is a positive constant. Find $\dfrac{d^2y}{dx^2}$ when $x = 2$ giving your answer in terms of k.

1.32 — Question 20. Prerequisite knowledge for Q20–Q27: Differentiation of the form ae^{kx} Evaluate $f'(1)$ for each of the following, leaving your answers in exact form:

a) $f(x) = e^{6x}$

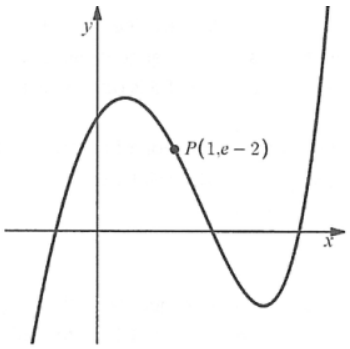
b) $f(x) = \sqrt{x} - e^{2x}$

c) $f(x) = x^{4/3} + e^{2x}$

1.32 — Question 21. Find the equation of the tangent line to the curve $y = e^x$ at $(0, 1)$ in the form $y = mx + c$ where m and c are integers to be found.

1.32 — Question 22. Find the equation of the normal line to the curve $y = x^2 - e^{2x}$ at $(0, -1)$ in the form $y = mx + c$ where m and c are integers to be found.

1.32 — Question 23. The curve of $y = e^x - 2x^2$ is shown on the diagram below. The point P has coordinates $(1, e - 2)$.



a) Find the equation of the tangent line at P, giving your answer in the form $y = mx + c$ where m and c are constants to be found.
b) Find the coordinates where the tangent line crosses the x-axis and y-axis.
c) Hence calculate the area of the triangle bounded by the tangent line at P and the coordinate axes giving your answer in the form $\dfrac{a}{b - e}$ where a and b are integers to find.

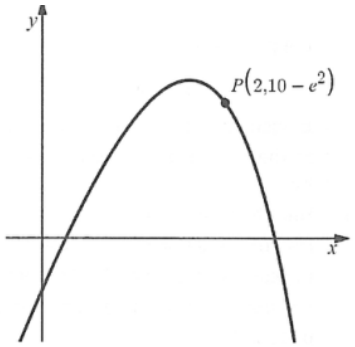
1.32 — Question 24. A curve has equation $y = 4e^{-kx}$ where k is a positive constant.

a) Find $\frac{dy}{dx}$ when $x = 2$ giving your answer in terms of k.

b) Show that $\frac{d^2y}{dx^2} = k^2y$.

1.32 — Question 25. A curve has equation $y = e^{5kx}(1 - k)$ where k is a positive constant. Given that at $x = \frac{1}{k}$, $\frac{d^2y}{dx^2} = -100e^5$, find the value of k.

1.32 — Question 26. The curve of $y = 5x - e^x$ is shown on the diagram below. The point P has coordinates $(2, 10 - e^2)$.

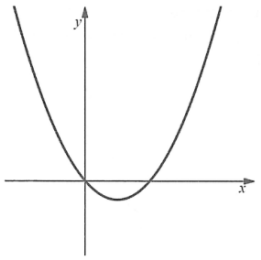


a) Find the equation of the normal line at P, giving your answer in the form $y = mx + c$ where m and c are constants to be found.
b) Find the coordinates where the normal line meet the x-axis, giving your answer in the form $(e^4 + ae^2 + b, 0)$ where a and b are integers to be found.

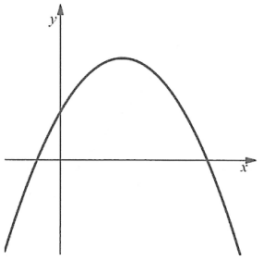
1.32 — Question 27. The normal to the curve $f(x) = e^x - 2x^2 - x + e^{-x}$ at $x = 0$ is the same line as the tangent to the curve $g(x) = \frac{1}{2}x^2 + (5 + k)x + p$ at the point $(7, 9)$. Find the values of p and k.

1.33 — Question 1. Draw the gradient function on copies of each of the following graphs:

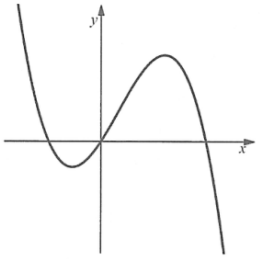
a)



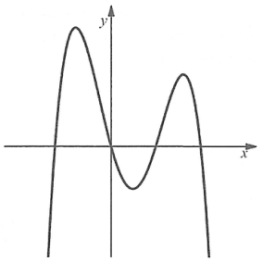
b)



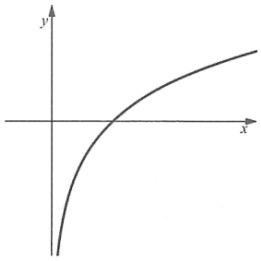
c)



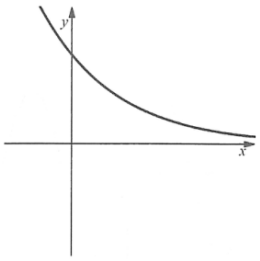
d)



e)

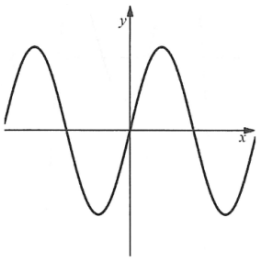


f)

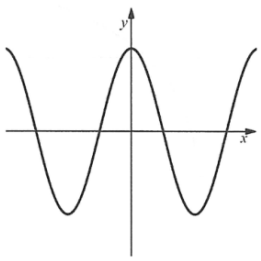


1.33 — Question 2. Given that in each of the following functions, their maximum and minimum gradient is the same magnitude as the maximum and minimum values respectively, draw the gradient function on copies of each of the following graphs:

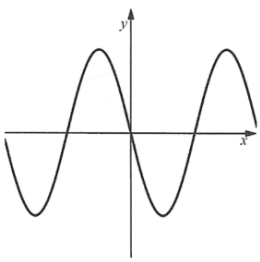
a)



b)



c)

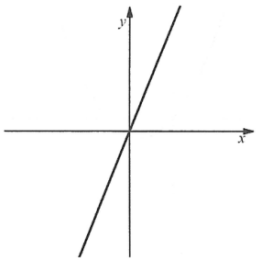


1.33 — Question 3. The gradient function of the curve

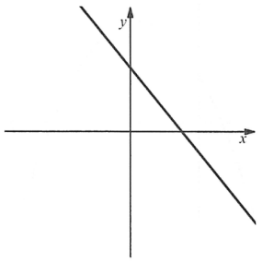
$$f(x) = ax^3 + bx^2 + cx$$

is sketched.
In each case state whether each of a, b, and c are greater than, equal to or less than 0

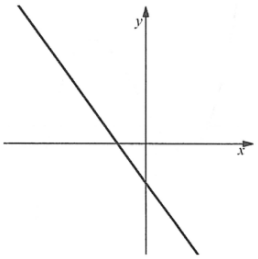
a)



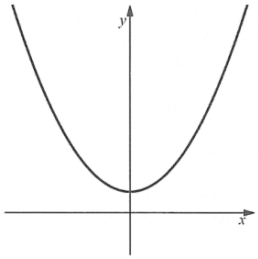
b)



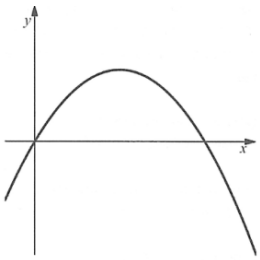
c)



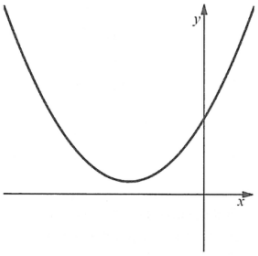
d)



e)



f)

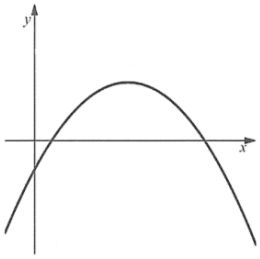


1.33 — Question 3 (continued). The gradient function of the curve

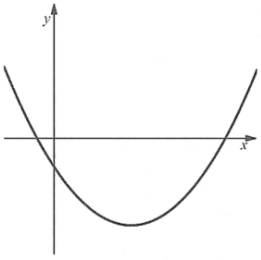
$$f(x) = ax^3 + bx^2 + cx$$

is sketched.
In each case state whether each of a, b, and c are greater than, equal to or less than 0

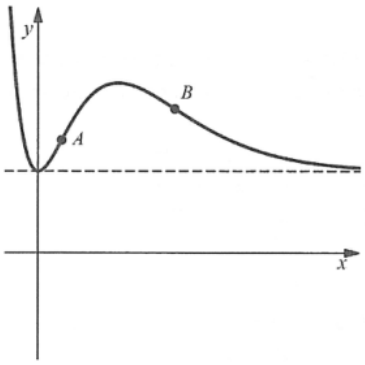
g)



h)



1.33 — Question 4. For the curve shown on the graph below, the two points of inflection are labelled A and B. Sketch the gradient function on a copy of the graph.



1.34 — Question 1. For each of following, find the value(s) of x for which $\frac{dy}{dx} = 0$.

a) $y = x^2 - 2x$

b) $y = -2x^2 + 18x - 37$

c) $y = x^3 - 3x^2 + 3x - 4$

d) $y = x - \frac{4}{x^2}$

e) $y = x + \frac{4}{x}$

f) $y = \frac{x^3 + 2}{x}$

g) $y = \frac{x^4 + 3}{3x^3}$

1.34 — Question 2. Find the interval on which each of the following functions are decreasing, giving your answer in set notation:

a) $f(x) = x^2 + 4x$

b) $f(x) = 6x^2 - 10x + 5$

c) $f(x) = x^3 - 3x$

d) $f(x) = 2x^3 + 3x^2 - 36x - 10$

e) $f(x) = -2x^3 - 18x^2 - 48x - 1$

1.34 — Question 3. Find the interval on which the function

$$f(x) = -x^3 - 6x^2 + 15x + 1$$

is increasing, giving your answer in interval notation.

1.34 — Question 4. Show that the function

$$f(x) = x^3 - 3x^2 + 4x - 8$$

is always increasing.

1.34 — Question 5. Show that the function

$$f(x) = -3x^3 + 3x^2 - 8x + 2$$

is always decreasing.

1.34 — Question 6. Find the coordinates of the stationary points of the following curves and determine their types:

a) $y = -x^2 + 5x - 6$

b) $y = 2x^3 - 15x^2 + 24x$

c) $y = 2x^3 - 3x^2 - 12x + 7$

d) $y = -x^3 + 3x^2 + 72x$

e) $y = 2x^4 - 4x^2$

f) $y = 4x + \dfrac{1}{x}$

g) $y = \dfrac{1}{x^2} + 2x$

h) $y = 3x^6 - 6x^3$

1.34 — Question 7. Find the coordinates of the maximum of the gradient function of

$$y = -3x^3 - 2x^2 + 1.$$

1.34 — Question 8. Find the coordinates of the stationary points, determine their types and hence sketch each of the following curves:

a) $y = x^3 - 3x^2 - 45x$

b) $y = 16x^4 - 32x^2 + 7$

c) $y = 5x - 2x^{\frac{1}{2}}$ for $x \geq 0$

1.34 — Question 9. Find the values of a, b and c such that

$$y = -x^3 + ax^2 + bx + c$$

has stationary points when $x = 1$ and $x = 3$, and passes through the point $(2, 4)$.

1.34 — Question 10. Find the values of a, b and c such that

$$y = ax^4 + bx^2 + c$$

passes through the points $(1, 0)$ and $(0, -3)$, and has a stationary point at $(\sqrt{2}, 1)$. Find all the stationary points and determine their types.

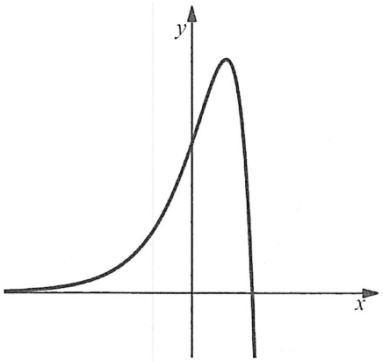
1.34 — Question 11.

Prerequisite knowledge for Q11–Q13: Differentiation of the form ae^{kx}

Giving your answer in set notation, find the interval on which $f(x) = e^{2x} - x$ is increasing.

1.34 — Question 12. The diagram shows the curve

$$y = 5e^x - e^{2x}.$$



a) Find the coordinates at which the curve crosses the x-axis and the y-axis.

b) Find the coordinates of the stationary point, giving your answer in exact form.

c) Show that the stationary point is a maximum.

1.34 — Question 13. Find the coordinates of the stationary point of

$$y = e^{3x} - e^{2x} - e^x$$

and determine its type.

1.35 — Question 1. For each of the following shapes, write an algebraic expression for the area, A and the perimeter, P:

a) Square with sides of length x

b) Rectangle with sides of length x and $2x$

c) Circle with radius x

d) Isosceles triangle with base of x and perpendicular height of $2x$

1.35 — Question 2. For each of the following 3D shapes, write an algebraic expression for the volume, V and the total surface area, A:

a) Cube with sides of length x

b) Cuboid with sides of length x, x and $3x$

c) Cylinder with radius x and a length h

d) Isosceles triangular based prism of length h, with the triangle surface having base x and a perpendicular height of x

1.35 — Question 3. For each of the following, find:

a) $\frac{dV}{dr}$ for $V = 20r - \frac{1}{4}r^2$

b) $\frac{dP}{dt}$ for $P = \frac{1}{t} + \frac{t}{2}$

c) $\frac{dA}{dr}$ for $A = 2r - \frac{10}{r^2}$

d) $\frac{dC}{dx}$ for $C = \frac{10}{x} + \frac{x^2}{4}$

1.35 — Question 4. Solve each of the following equations, leaving your answer in exact form:

a) $25\pi - 4\pi r^2 = 0$, $r > 0$

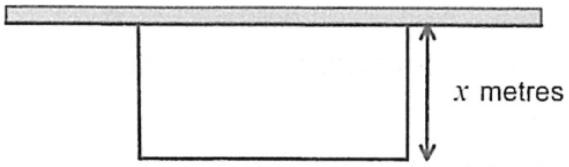
b) $-\frac{144}{x^2} + \frac{1}{4} = 0$, $x > 0$

c) $-\frac{10}{x^2} + \frac{1}{2}x = 0$, $x > 0$

d) $\frac{1}{4}\pi r - \frac{27\pi^2}{r^2} = 0$, $r > 0$

e) $\frac{1}{5}\pi r - \frac{3}{r^2} = 0$, $r > 0$

1.35 — Question 5. A rectangular enclosure is created within a field, with a wall forming one side with a width of x metres. A rope of length 30 metres is used to form the remaining three sides.



a) Show that the area A may be written as $A = 30x - 2x^2$.

b) Hence, find the maximum area of the enclosure.

1.35 — Question 6. The cost of a journey £C when a van when driving at a steady speed in v kilometres per hour may be modelled by

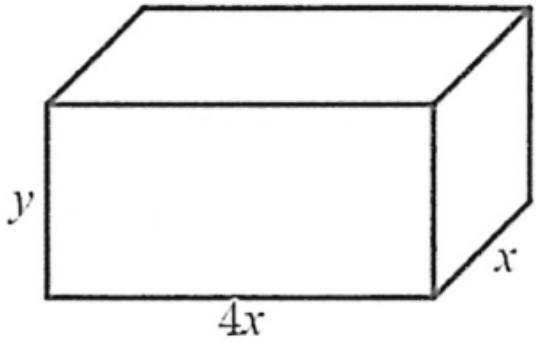
$$C = \frac{1156}{v} + \frac{v}{9} + 50$$

where $v > 0$.

a) Find the speed at which the minimum cost occurs.

b) Find the minimum cost.

1.35 — Question 7. The diagram shows a solid cuboid with lengths x cm, $4x$ cm and y cm.



The total surface area of the cuboid is 100 cm^2.

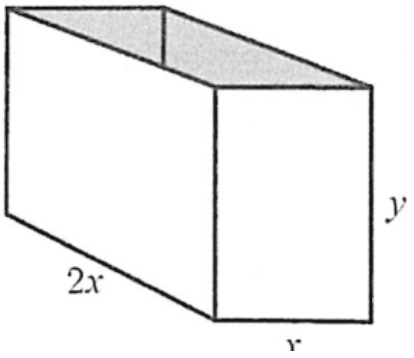
a) Show that the volume may be written as

$$V = 40x - \frac{16}{5}x^3.$$

b) Hence, find the maximum volume of the cuboid giving your answer to 1 decimal place.

c) Find the exact dimensions of the box for this maximum volume to occur.

1.35 — Question 8. The diagram shows an open-topped cuboid tank. Its base has width x metres and length $2x$ metres and has a height of y metres.



It is made from 48 m^2 of sheet metal.

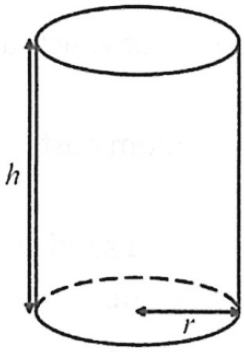
a) Show that the volume may be written as

$$V = 16x - \frac{2}{3}x^3.$$

b) Hence, find the maximum volume giving your answer to 1 decimal place.

c) Find the exact dimensions of the box for this maximum volume to occur.

1.35 — Question 9. The diagram shows a plant pot which is a cylindrical container of radius r cm and height h cm. The plant pot has an open top. The total surface area is 48π cm^2.

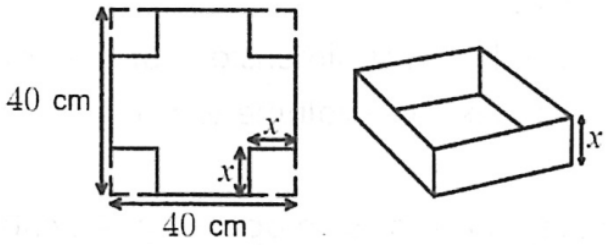


a) Show that the volume may be written as

$$V = 24\pi r - \frac{\pi}{2}r^3.$$

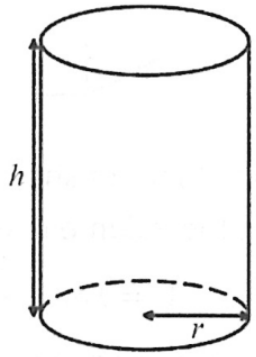
b) Hence find the maximum volume giving your answer in exact form.

1.35 — Question 10. The diagram shows a piece of square cardboard with width 40 cm. Four squares are cut from the corners each of width x cm. The cardboard is then folded into an open tray.



Find the maximum capacity of the tray. Give your answer to 1 decimal place.

1.35 — Question 11. A container is created as a closed cylinder with capacity 432π cm^3.



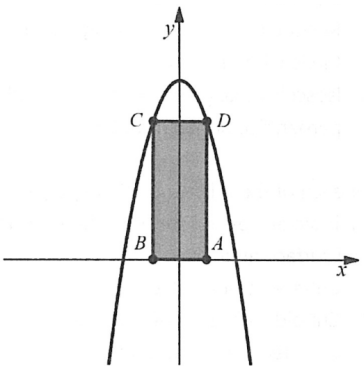
a) Find the dimensions of the container when its surface area is minimised.

b) For sealing the side of the container, a line of fluid is dropped along the length. Find the cost given it costs £0.04 per cm.

1.35 — Question 12. The curve

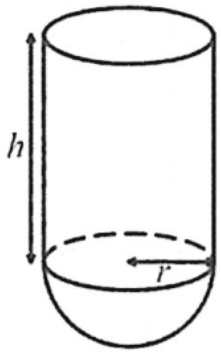
$$f(x) = 10 - x^2$$

and rectangle ABCD is shown in the diagram below.



Given that the rectangle is bounded by the curve and the x-axis, find the maximum area that the rectangle can have.

1.35 — Question 13. A container for ice cream desserts is being designed to use as little plastic as possible. It needs to have a volume of 144π cm^3. The vendors of the ice cream would like it to be comprised of an open top semi-sphere with a cylinder with the same radius attached, and that it should have no base or top. The radius of the cylinder is r cm and the length of the cylinder is h cm.



A sphere of radius r has volume $\frac{4}{3}\pi r^3$ and surface area $4\pi r^2$.

a) Show that the area of plastic used is given by

$$A = \frac{2}{3}\pi r^2 + \frac{288\pi}{r}.$$

b) Hence find the radius which minimises the area.

c) By calculating the value of h for this to be minimised, state an issue with the vendor's requirement.

1.35 — Question 14. A cylindrical metal container which is closed at either end, has a capacity of 500 cm^3. The metal used for the curved side costs 0.03 pence/cm^2. The cost for the circular bottom and top costs 0.07 pence/cm^2.

a) Show that the cost of the metals for making the container may be given by

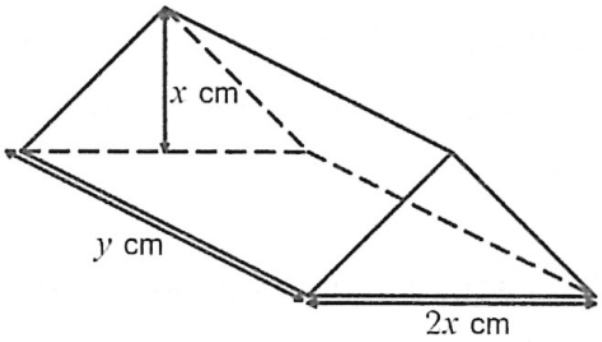
$$C = \frac{7}{50}\pi r^2 + \frac{30}{r}$$

where r is the radius of the container and C is the cost in pence.

b) Find the radius for the minimum cost and height for that cost.

c) Hence calculate the minimum cost of making one of these containers to the nearest pence.

1.35 — Question 15. A solid triangular based prism is made from wood with a volume of $\left(500(\sqrt{2}-1)\right)$ cm^3. The two triangular ends are isosceles triangles with base $2x$ cm and perpendicular height x cm. The length of the prism is y cm. The rectangular sides are painted with paint that costs £0.02 per cm^2 and the triangular ends are painted with paint that costs £0.04 per cm^2.



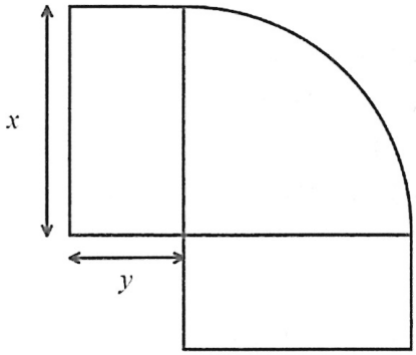
a) Show that the slanted sides (not the base) have a total area of $2\sqrt{2}xy$.

b) Show that the cost of painting the prism is given by

$$C = \frac{20}{x} + \frac{2}{25}x^2.$$

c) Hence calculate the minimum cost of painting one of these prisms.

1.35 — Question 16. A part of a garden is going to be filled with two different types of decorative stones. The design is a quarter circle with radius x. Each rectangle has width y metres and length x metres and is laid out as shown in the diagram.



1.35 — Question 16 (continued). For the quarter circle, the stones will be 80% of the cost of the stones for the rectangular areas. The total perimeter is 16 metres.

a) Show that the cost may be given by

$$C = 8px - \left(1 + \frac{\pi}{20}\right) px^2$$

for some cost £p per metre^2.

b) Show that the length of x for the maximum cost is occurs when

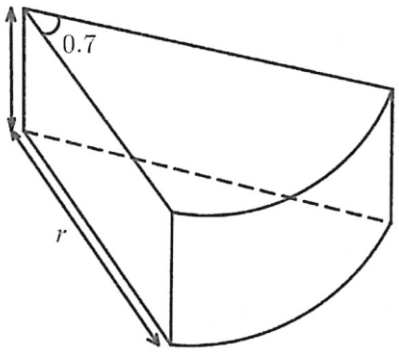
$$x = \frac{80}{20 + \pi}.$$

c) Calculate the area of the quarter circle and the rectangles separately for this value of x, giving your answer to 3 significant figures.

1.35 — Question 17.

Prerequisite knowledge for Q17: Arc Length and Sector Area in Radians

A toy slice of cake is being manufactured and can be modelled by a sector of a circle with radius r. The volume is to be 350 cm^3. The angle of the sector is 0.7 radians.



a) Show that the surface area may be written in the form

$$A = \frac{2700}{r} + 0.7r^2.$$

b) Hence find the minimum surface area and the values of r and h which obtain this, giving your answer to 3 significant figures.